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Acoustic Damper Placement and Tuning for Annular Combustors: An Adjoint-Based Optimization Study

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Thermoacoustic instabilities pose a major threat to modern gas turbines. The use of acoustic dampers, like Helmholtz resonators, has proven useful for the mitigation of such instabilities. However, assessing the effect of acoustic dampers on thermoacoustic modes in annular combustion chambers remains an intricate task. This results from the implicit nature of the thermoacoustic Helmholtz equation associated with the high number of possible parameter values for the positioning of the dampers and their impedance design. In the present work, the principal challenges of the effective placement and the design of the impedance of acoustic dampers in annular chambers are discussed. This includes the choice of an appropriate objective function for the optimization, the combinatorial challenges when dealing with different possible damper arrangements, and the numerical complexities when using the thermoacoustic Helmholtz equation to approach this issue. As a key aspect, the paper proposes a new adjoint-based approach to tackle these problems. The new algorithm establishes algebraic models that predict the effect of acoustic dampers on the growth rates of the thermoacoustic modes. The theory is exemplified on the basis of a generic annular combustor model with 12 burners.
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Introduction

Thermoacoustic instabilities pose a major threat to modern gas turbines. These instabilities have negative effects on the operational range of the combustors and may severely damage the machines [1]. Both passive and active damping methods may be considered to tackle this issue. Usually, passive methods are used because they feature high reliability at low costs. Such methods are based on the use of acoustic damping devices like Helmholtz dampers. It is very difficult to adapt these components after machine installation. Thus, the availability of fast and accurate mathematical design tools is essential for the development of new gas turbines featuring a higher efficiency and greater lifetime.

Large eddy simulations may be considered for the assessment of thermoacoustic instabilities [2–4], but their numerical costs are too high to be used for design purposes. As an alternative, the governing equations can be linearized and transformed to the frequency domain. Thermoacoustic instabilities can then be modeled using a thermoacoustic Helmholtz equation:

$$\nabla \cdot (c^2 \nabla \hat{p}) + \omega^2 \hat{p} = -\mathrm{i}\omega(\gamma - 1)\hat{q} \tag{1}$$

Here, c denotes the local speed of sound, $\hat{p}$ is the Fourier transform of the pressure fluctuation; ω is the complex frequency of oscillation, γ is the ratio of specific heats, and $\hat{q}$ is the Fourier transform of the volumetric unsteady heat release rate. To close the problem, the heat release rate can be related to the velocity fluctuation at a reference point by means of a flame transfer function. Finally, this allows the heat release rate to be related to the acoustic field by a linear differential operator $\mathcal{Q}$.

$$-\mathrm{i}\omega(\gamma - 1)\hat{q} = \mathcal{Q}\hat{p} \tag{2}$$

The linear operator $\mathcal{Q}$ evaluates the axial velocity fluctuation at a reference position as axial component of the gradient of the pressure fluctuation. It also accounts for a time lag and the spatial distribution of the heat release zone.

Finally, the Helmholtz equation takes the form of a nonlinear eigenvalue problem with eigenvalue ω and eigenfunction $\hat{p}$. For convenience, the problem can be written as

$$\mathcal{L}(\omega)\hat{p} = 0 \tag{3}$$

where

$$\mathcal{L}(\omega)\hat{p} := \nabla \cdot (c^2 \nabla \hat{p}) + \omega^2 \hat{p} - \mathcal{Q}\hat{p} \tag{4}$$

To assess the stability of a mode, the imaginary part of the eigenfrequency ω must be evaluated. It is possible to obtain ω by solving the above eigenvalue problem utilizing a discretization method. However, the implicit nature of the above eigenvalue problem obscures the effect of changes to some model parameters on the stability. This means that the equation must be solved again for each new parameter set. This is a numerically costly task, as already one solution process requires several fixed-point iterations involving repeated solutions of large linear eigenvalue problems [5].

This is especially a problem because acoustic damping devices are modeled in terms of frequency-dependent impedance boundary conditions, the behavior of which again depends on several design parameters.

$$cZ(\boldsymbol{\zeta})\nabla \hat{p} \cdot \mathbf{n} + \mathrm{i}\omega\hat{p} = 0 \tag{5}$$

Here, Z denotes the reduced impedance, i.e., the impedance normalized by the characteristic impedance ρc of the considered fluid. n refers to the outward pointing unit vector normal to the impedance boundary and $\boldsymbol{\zeta}$ is the vector of impedance design parameters. For instance, the impedance of a Helmholtz damper is affected by its neck length, its volume, the neck cross section, and

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the gas temperature [6]. In addition, many possible positioning patterns have to be considered [7] when other design constraints limit the number of damping devices. Altogether, this means that finding the optimum positioning pattern and best possible tuning for all applied acoustic dampers is a task involving a high number of relevant parameters. Brute force testing of all possible parameter configurations is therefore not feasible. This is sometimes referred to as the curse of dimensionality.

Lumped network models are commonly used as a discretization method to overcome this issue. Dowling and Stow used this technique to study a generic geometry and derive rules for the best positioning of acoustic dampers [7]. Bothien et al. combined network models with a genetic optimization algorithm to optimize a new damper design [8]. Recently, Schuermans et al. used the same approach for the design of a novel damping–cooling system for Alstom's GT26/GT24 engines [9]. However, the numerical efficiency of the network models arises from a significant geometrical reduction of the originally three-dimensional geometry to a network of one-dimensional surrogate components, if the model is not coupled to a spatial discretization method like finite elements.

Full domain discretization methods may therefore be considered, in order to maintain the geometrical complexity of the actual system in the model representation. Camporeale et al. used a finite element method (FEM) to study the effect of multiple Helmholtz resonators equipped to an annular combustion chamber [6]. They observed that the application of Helmholtz dampers splits the resonant mode into two modes with distinct frequencies. This can be attributed to the loss of discrete rotational symmetry. However, this approach is numerically too complex to be used as fast design tool.

Adjoint perturbation theory offers an opportunity to overcome this problem. It approximates the change in the eigenfrequency and the eigenfunction of nonlinear, nonhermitian eigenvalue problems due to changes in the involved operators. The following problem is considered:

If ω_0 and $\hat{p}_0$ is a known eigenpair of the operator family $\mathcal{L}_0$, how does it change when the family is altered to

$$\mathcal{L}(\omega) = \mathcal{L}_0(\omega) + \varepsilon\mathcal{L}_1(\omega) \tag{6}$$

by the perturbation $\varepsilon\mathcal{L}_1(\omega)$. Power series ansätze are established to facilitate the approximation. This results in explicit expressions for both the eigenvalue and the eigenfunction which are much faster to evaluate than the implicit eigenvalue problem.

In the remainder of this paper, it will be discussed how adjoint perturbation theory can be utilized to (i) find the best feasible positioning pattern for a set of acoustic dampers and (ii) perform a local optimization within the space of damper parameters. The discussion starts with a brief summary of the mathematical fundamentals of the theory and a detailed outline of how adjoint perturbation theory can be tailored into an optimization algorithm. Subsequently, the application of the theory to a generic annular combustion chamber model is discussed. The paper closes with remarks on how to further improve the presented theory and propositions for other applications.

Adjoint Perturbation Theory

This section reports the mathematical fundamentals for the optimization theory. A detailed discussion of the basic concepts is beyond the scope of this paper. However, the mentioned references are valuable sources for the interested reader. We refer to the theory as adjoint perturbation theory, though other notions are widely used. The theory has its roots in quantum mechanics where it is referred to as spectral perturbation theory [10], or Rayleigh–Schrödinger perturbation theory [11]. The lack of self-adjointness of the thermoacoustic Helmholtz equation makes it necessary to include adjoints of the involved operators. The theory is then also called adjoint sensitivity analysis [12]. However, this notion is used by us for the first-order theory only. A nice review of adjoint equations with focus on hydrodynamic instabilities can be found in Ref. [13].

The key idea behind adjoint perturbation theory is to establish asymptotic power series ansätze for both the eigenvalue and the eigenvector in order to find approximations to the solutions of the perturbed nonlinear eigenvalue problem (6)

$$\hat{p} = \hat{p}_0 + \sum_{n=1}^{N} \varepsilon^n \hat{p}_n = \sum_{n=0}^{N} \varepsilon^n \hat{p}_n \tag{7}$$

$$\omega = \omega_0 + \sum_{n=1}^{N} \varepsilon^n \omega_n = \sum_{n=0}^{N} \varepsilon^n \omega_n \tag{8}$$

The nonlinearity of the problem requires a Taylor-series expansion of the operators $\mathcal{L}_0(\omega)$ and $\mathcal{L}_1(\omega)$

$$\mathcal{L}_0(\omega_0 + \Delta\omega) \approx \mathcal{L}_0(\omega_0) + \sum_{m=1}^{N} \frac{1}{m!} (\Delta\omega)^m \mathcal{L}_0^{\{m\}}(\omega_0) \tag{9}$$

$$\mathcal{L}_1(\omega_0 + \Delta\omega) \approx \mathcal{L}_1(\omega_0) + \sum_{m=1}^{N} \frac{1}{m!} (\Delta\omega)^m \mathcal{L}_1^{\{m\}}(\omega_0) \tag{10}$$

Here, the superscript $\{m\}$ denotes the mth derivative with respect to ω. Using these series expansions and inserting the ansätze (7) and (8), as well as Eq. (6), into Eq. (3) yields after sorting like powers of ε and considering the linear independence of these powers

$$\mathcal{L}_0(\omega_0)\hat{p}_k = -r_k - \omega_k \mathcal{L}_0^{\{1\}}(\omega_0)\hat{p}_0 \tag{11}$$

where

$$r_k = \mathcal{L}_1(\omega_0)\hat{p}_{k-1} + \sum_{m=1}^{N} \frac{1}{m!} \sum_{\substack{|\mu|=m \\ \mu \neq 1_k}} \binom{m}{\mu} \omega_\nu^\mu \left(\mathcal{L}_0^{\{m\}}(\omega_0)\hat{p}_{k-|\nu\mu|} + \mathcal{L}_1^{\{m\}}(\omega_0)\hat{p}_{k-1-|\nu\mu|} \right) \tag{12}$$

Here, μ denotes a multiindex of length N; $\binom{m}{\mu}$, a multinomial coefficient; and ν, the multiindex having the entries $(1, 2, \ldots, N)$. 1_k denotes a multiindex where all entries except for the kth entry are zero, and at the kth position, it is one. The notation ω_ν^μ abbreviates the product $\prod_{n=1}^{N} \omega_n^{\mu_n} = \omega_1^{\mu_1}\omega_2^{\mu_2}\cdots\omega_N^{\mu_N}$. Please note the convention that entities with negative subscripts equal the zero elements of their respective sets.

For instance, the equations for order one and two are

$$\mathcal{L}_0(\omega_0)\hat{p}_1 = -\underbrace{\mathcal{L}_1(\omega_0)\hat{p}_0}_{r_1} - \omega_1 \mathcal{L}_0^{\{1\}}(\omega_0)\hat{p}_0 \tag{13}$$

$$\mathcal{L}_0(\omega_0)\hat{p}_2 = \tag{14}$$

$$-\left(\underbrace{\mathcal{L}_1(\omega_0)\hat{p}_1 + \omega_1\left(\mathcal{L}_0^{\{1\}}(\omega_0)\hat{p}_1 + \mathcal{L}_1^{\{1\}}(\omega_0)\hat{p}_0\right) + \frac{1}{2}\omega_1^2 \mathcal{L}_0^{\{2\}}(\omega_0)\hat{p}_0}_{r_2} \right) - \omega_2 \mathcal{L}_0^{\{1\}}(\omega_0)\hat{p}_0 \tag{15}$$

Equation (11) gives rise to a determination algorithm for the corrections of the eigenvalues and eigenfunctions: Provided that the eigenfunction and eigenvalue corrections up to order $k-1$ are already known, the only unknowns in this equation are the kth-order corrections $\hat{p}_k$ and ω_k, which themselves only occur on the

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left- and right-hand side, respectively. Taking the inner product of the adjoint eigenvector $\hat{p}_0^\dagger$ with the right-hand side yields

$$\left\langle \hat{p}_0^\dagger \middle| \mathcal{L}_0(\omega_0)\hat{p}_k \right\rangle = \left\langle \mathcal{L}_0^\dagger(\omega_0)\hat{p}_0^\dagger \middle| \hat{p}_k \right\rangle = 0 \tag{16}$$

Hence, the inner product with the left-hand side must also vanish and a determination equation for ω_k can be derived from Eq. (11) by taking the inner product with $\hat{p}_0^\dagger$. Considering that the eigenfunctions are possibly D-fold degenerate, this rationale leads to the following generalized algebraic eigenvalue problem:

$$\mathbf{Xz} + \omega_k \mathbf{Yz} = 0 \tag{17}$$

where $[\mathbf{X}]_{ba} = \langle \hat{p}_{0,b}^\dagger | r_{k,a} \rangle$, $[\mathbf{Y}]_{ba} = \langle \hat{p}_{0,b}^\dagger | \mathcal{L}_0^{\{1\}}(\omega_0)\hat{p}_{0,a} \rangle$ and $\mathbf{z} = [z_1 \quad \dots \quad z_D]$.

This eigenvalue problem should not be confused with the nonlinear eigenvalue problem given through the Helmholtz equation (3). It is an auxiliary $D \times D$ eigenvalue problem occurring when mode degeneracy is encountered. The D eigenvectors $\mathbf{z}$ then determine the coefficients for the linear combinations $\sum_{i=1}^{D} z_i \hat{p}_{0,i}$ which describe the subspaces to which the degenerate eigenspace unfolds. The eigenvalues ω_k appearing here specify the corresponding kth-order eigenvalue corrections ω_k for the power series (8). The eigenvalue problem (17) is therefore called the mode unfolding problem. The corresponding eigenfunction corrections can be found by substituting ω_k back into Eq. (11). Solvability of this equation is guaranteed by virtue of Eq. (16) [14].

The discrete rotational symmetry of annular combustors is a trigger of two-fold mode degeneracy for most of its thermoacoustic modes. The application of dampers may break this symmetry and lead to two modes oscillating at different eigenfrequencies close to the original degenerate frequency [7,15]. This makes the correct computation of the mode unfolding highly relevant for the current context.

The Role of Epsilon

Some authors consider the perturbation parameter ε as rather conceptual. It is therefore set to one after the derivation of the equations. However, it can be linked to physical parameters [10]. In this paper, these parameters are the design parameters $\boldsymbol{\zeta}$ for the acoustic dampers. This offers the opportunity to interpret the power series ansätze (7) and (8) as functions describing the influence of the physical parameters on the mode shape and frequency. The accuracy of these functions is limited due to their asymptotic nature and hence a method for the estimation of the approximation error is needed. This error η is defined as deviation of the correct eigenvalue from the eigenvalue prediction:

$$\eta_N = \omega - \sum_{n=0}^{N} \varepsilon^n \omega_n \tag{18}$$

A common approach for the estimation of this error would be assuming that it is of the same order as the first neglected term in the series. There is no proof that this truncation rule applies in the current context, but various examples from the field of quantum mechanics involving the Schrödinger equation, which is a wave equation as well, are known, see, e.g., Ref. [10]. Moreover, there is a variety of mathematical and scientific examples for perturbations where the truncation rule applies [16]. We therefore set up a numerical study investigating a Rijke tube with a variable impedance at the outlet. This study will demonstrate that the truncation rule applies for the investigation of thermoacoustic modes and hence facilitate a fast estimation of the perturbation error. In this study, a simple two-dimensional Rijke tube was considered. An n-τ-model was used to model the flame response. The cold gas temperature was set to $T_u = 300\,\mathrm{K}$ while the hot gas temperature was set to $T_b = 1500\,\mathrm{K}$. The ratio of specific heats was set to $\gamma = 1.4$. The geometry is depicted in Fig. 1. The interaction index was set

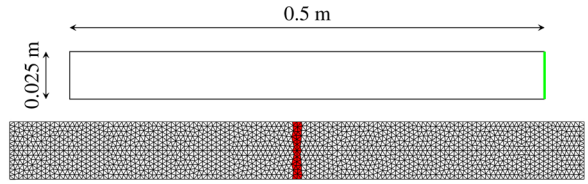

Fig. 1 Considered test geometry and discretization mesh. The black boundaries were assumed to be sound hard while a variable impedance could be specified at the outlet (green). The red shaded elements highlight the flame volume.

to $n = 0.1$ and the time delay to $\tau = 2$ ms. The reference point for the flame transfer function was located in the middle of the tube 0.01 m upstream of the flame.

A finite element method utilizing triangular linear Lagrange elements was employed to discretize the thermoacoustic Helmholtz equation. This led to the following eigenvalue problem:

$$\underbrace{\left[\mathbf{K} + \omega\frac{1}{Z_0}\mathbf{B} + \omega^2\mathbf{M} - n\exp(-\mathrm{i}\tau\omega)\mathbf{Q}\right]}_{\mathcal{L}_0(\omega)}\mathbf{p} + \underbrace{\left[\frac{1}{Z_1} - \frac{1}{Z_0}\right]}_{\varepsilon}\underbrace{\omega\mathbf{B}}_{\mathcal{L}_1(\omega)}\mathbf{p} = 0 \tag{19}$$

Here, $\mathbf{K}$ denotes the stiffness matrix, $\mathbf{B}$ is the boundary mass matrix at the outlet, $\mathbf{M}$ is the mass matrix, $\mathbf{Q}$ is the discretization matrix originating from the heat release operator, and $\mathbf{p}$ is the discretized pressure fluctuation amplitude. Z_0 is the unperturbed impedance while Z_1 refers to the perturbed impedance. With this formulation, the effect on the eigenfrequency when changing the impedance from Z_0 to Z_1 can be computed utilizing adjoint perturbation theory, with the difference of the admittances being the perturbation parameter. In the unperturbed case, the reduced admittance at the outlet was set to $(1/Z_0) = 8 - 4\mathrm{i}$. In a first test, the perturbation was set to $\epsilon = (1/Z_1) - (1/Z_0) = 0.8 - 1.6\mathrm{i}$. The effect of this admittance change on the eigenfrequency was computed using both a classical solution of the new eigenvalue problem and the perturbation theory to calculate the result from the unperturbed solution. The eigenvalues for two, four, five, and eight times amplified perturbations were also computed. Please note that the correction terms for these cases can be easily obtained by rescaling the correction terms of the original perturbation with appropriate powers of the amplification σ:

$$\omega = \sum_{n=1}^{N} (\sigma\epsilon)^n \omega_n = \sum_{n=1}^{N} \sigma^n \epsilon^n \omega_n \tag{20}$$

This study allows to asses the accuracy of the perturbation method in terms of the error. Figure 2 shows the results of this study for the mode which has an unperturbed eigenfrequency of $\omega/2\pi = 225.6\,\mathrm{Hz} + 7.2\mathrm{i}\,\mathrm{s}^{-1}$. All computations were conducted using MATLAB. The magnitude of the error $|\eta_N|$ and the error of the first neglected term of the power series for the eigenfrequency correction $|\sigma^{N+1}\epsilon^{N+1}\omega_{N+1}|$ are compared.

The correspondence between the magnitude of the first neglected term and the error for errors greater than 10^{-10} Hz can be clearly seen. The deviation for smaller magnitudes may be attributed to the machine precision. However, such accuracy is not relevant for practical applications. Seemingly, the error decreases for perturbation strength of 1, 2, and 4 when the applied perturbation order N is increased. On the contrary, the error does not change for a strength of five and even increases for a perturbation strength of eight. This may be because the convergence radius of the power series might be zero so that the series is divergent. The convergence radius is actually independent of the perturbation strength. However, the smaller the perturbation strength

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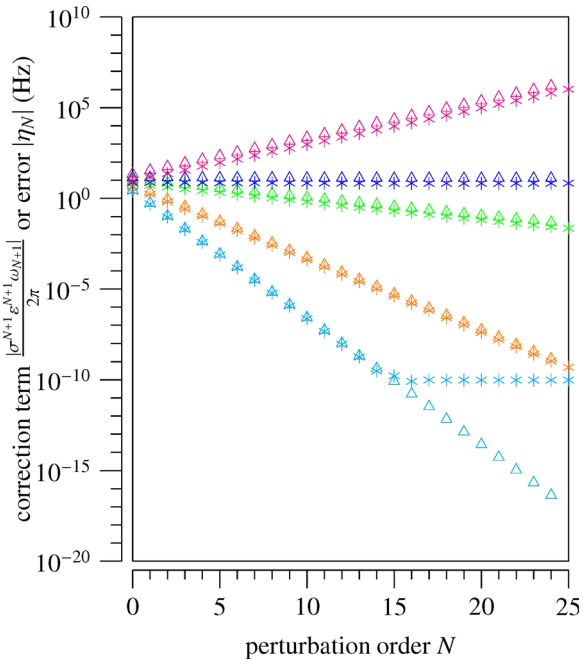

Fig. 2 Comparison of the first neglected term $\sigma^{N+1}\epsilon^{N+1}\omega_{N+1}$ and the corresponding error η_N. The first neglected term is indicated by a triangle while the error is marked by an asterisk. The different colors cyan, orange, green, blue, and magenta correspond to the perturbation amplification $\sigma = 1, 2, 4, 5,$ and 8, respectively.

is, the higher the applied perturbation order must be for this to become apparent. Nevertheless, in many cases if $N \to \infty$, then $\eta_N \to \infty$. Obviously, an order of $N = 25$ is not high enough to see this effect for all considered perturbation strengths. The fact that for small perturbations the error decreases at low perturbation orders before reaching a minimum and then increasing to infinity is a common feature of perturbation theory. It is known as superasymptotics [16].

Please note the slightly increasing offset between the first neglected term and the error for increasing perturbation strengths. For the current case, the magnitude of the first neglected term is larger than that of the error when machine precision is not limiting the calculation. Further considerations have to clarify whether this is a general feature or cases might exist where the first neglected term slightly underestimates the error. No matter whether the deviation overrates the error or not, it will scale with the perturbation strength. Hence, the offset increases with the perturbation strength. Thus, the method may fail to predict the error when the perturbation strength becomes very large. However, because the the error is also very large this is irrelevant for practical purposes.

The results indicate that the truncation rule holds true for the considered test case. This supports the hypothesis that the truncation rule applies for perturbations of the thermoacoustic Helmholtz equation. The results also emphasize that the error decreases with the perturbation strength. The truncation rule can be utilized to calculate the strength of a parameter shift such that the error is not greater than a specified tolerance. This is a crucial aspect in view of the possible divergence of the applied power series. Most importantly, it allows to limit the perturbation strength such that the predicted eigenvalue shift is highly accurate, a key idea for the iterative optimization algorithm proposed later.

First-Order Multiparameter Perturbations

The aforementioned theory is very useful in scenarios where the perturbation is known in advance. Then the perturbation can be formulated with one perturbation operator $\mathcal{L}'$ and one perturbation parameter ε at a specific value. As the positioning of multiple acoustic dampers involves a large number of relevant parameters, a multiparameter perturbation theory is necessary, i.e., a mathematical tool to approximate solutions of the following perturbation problem:

$$\mathcal{L}(\omega) = \mathcal{L}_0(\omega) + \sum_{p=1}^{P} \varepsilon_p \mathcal{L}_p(\omega) \tag{21}$$

Here, ε_p and $\mathcal{L}_p$ denote the perturbation parameters and perturbation operators describing the effect of changes to P independent design parameters. This section develops such a theory. For the present application, only a first-order multiparameter theory is needed and as higher-order multiparameter perturbations are a complicated matter if mode degeneracy occurs [11, p.116 ff.], the discussion is restricted to a first-order analysis.

Again power series ansätze may be established to tackle this issue

$$\omega = \omega_0 + \sum_{p=1}^{P} \omega_{1,p}\varepsilon_p \tag{22}$$

However, in case of mode degeneracy, the sensitivities will not be constant but functions of all independent parameters.

$$\omega_{1,p} = \omega_{1,p}(\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_P) \tag{23}$$

Substituting $\sum_{p=1}^{P} \varepsilon_p \mathcal{L}_p = \mathcal{L}'$ and $\sum_{p=1}^{P} \omega_{1,p}\varepsilon_p = \omega_1$, the multiparameter eigenvalue correction formula can be derived from the one-parameter correction formula

$$\left\langle \hat{p}_0^{\dagger} \middle| \mathcal{L}'(\omega_0)\hat{p}_0 \right\rangle + \omega_1 \left\langle \hat{p}_0^{\dagger} \middle| \mathcal{L}_0^{(1)}(\omega_0)\hat{p}_0 \right\rangle = 0 \tag{24}$$

In the nondegenerate case, this can be rearranged to

$$\left\langle \hat{p}_0^{\dagger} \middle| \sum_{p=1}^{P} \varepsilon_p \mathcal{L}_p(\omega_0)\hat{p}_0 \right\rangle + \sum_{p=1}^{P} \omega_{1,p}\varepsilon_p \left\langle \hat{p}_0^{\dagger} \middle| \mathcal{L}_0^{(1)}(\omega_0)\hat{p}_0 \right\rangle = 0 \tag{25}$$

$$\sum_{p=1}^{P} \varepsilon_p \left(\left\langle \hat{p}_0^{\dagger} \middle| \mathcal{L}_p(\omega_0)\hat{p}_0 \right\rangle + \omega_{1,p} \left\langle \hat{p}_0^{\dagger} \middle| \mathcal{L}_0^{(1)}(\omega_0) \right\rangle \right) = 0 \tag{26}$$

Here, the term $\langle \hat{p}_0^{\dagger} | \mathcal{L}_p(\omega_0)\hat{p}_0 \rangle + \omega_{1,p} \langle \hat{p}_0^{\dagger} | \mathcal{L}_0^{(1)}(\omega_0) \rangle$ is exactly the same as if the pth perturbation would be treated as a single perturbation. Thus, the first-order correction is just the superposition of the P eigenvalue corrections when those are individually applied. This is not surprising in view of a first-order theory. However, this simple superpositioning rule does not hold true for degenerate cases. This is because of the unfolding of the eigenspace. This can be seen when inserting the substitutions into the mode unfolding problem (17). This yields

$$\sum_{p=1}^{P} \varepsilon_p \mathbf{X}_p \mathbf{z} + \sum_{p=1}^{P} \varepsilon_p \omega_{1,p} \mathbf{Y}\mathbf{z} = 0 \tag{27}$$

where $[\mathbf{X}_p]_{ba} = \langle \hat{p}_{0,b} | \mathcal{L}_p(\omega_0)\hat{p}_{0,a} \rangle$, $[\mathbf{Y}]_{ba} = \langle \hat{p}_{0,b}^{\dagger} | \mathcal{L}_0^{(1)}(\omega_0)\hat{p}_{0,a} \rangle$.

It is possible to factor out $\sum_{p=1}^{P} \varepsilon_p$, but the terms $\mathbf{X}_p\mathbf{z} + \omega_{1,p}\mathbf{Y}\mathbf{z}$ define eigenvalue problems themselves (the mode unfolding problems for the respective single parameter perturbations) and superposition does in general not hold true for eigenvalue problems.

A loophole to this problem is provided by the fact that in annular combustors, the mode degeneracy, if there is any, is two-fold. Therefore, the eigenvalue problem (27) is given by a 2×2-system. This system can easily be solved analytically by equating

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its determinant to zero. The determinant is given by a quadratic polynomial in $\omega_1 = \sum_{p=1}^{P} \varepsilon_p \omega_{1,p}$, namely

$$\det\left(\sum_{p=1}^{P} \varepsilon_p \mathbf{X}_p + \underbrace{\sum_{p=1}^{P} \varepsilon_p \omega_{1,p}}_{=\omega_1} \mathbf{Y}\right) = a\omega_1^2 + b\omega_1 + c \tag{28}$$

where

$$a = [\mathbf{Y}]_{11}[\mathbf{Y}]_{22} - [\mathbf{Y}]_{12}[\mathbf{Y}]_{21} = \det(\mathbf{Y}) \tag{29}$$

$$b = [\mathbf{Y}]_{11} \sum_{p=1}^{P} \varepsilon_p [\mathbf{X}_p]_{22} + [\mathbf{Y}]_{22} \sum_{p=1}^{P} \varepsilon_p [\mathbf{X}_p]_{11}$$

$$-[\mathbf{Y}]_{21} \sum_{p=1}^{P} \varepsilon_p [\mathbf{X}_p]_{12} - [\mathbf{Y}]_{12} \sum_{p=1}^{P} \varepsilon_p [\mathbf{X}_p]_{21} \tag{30}$$

$$c = \left(\sum_{p=1}^{P} \varepsilon_p [\mathbf{X}_p]_{11}\right)\left(\sum_{p=1}^{P} \varepsilon_p [\mathbf{X}_p]_{22}\right) - \left(\sum_{p=1}^{P} \varepsilon_p [\mathbf{X}_p]_{12}\right)\left(\sum_{p=1}^{P} \varepsilon_p [\mathbf{X}_p]_{21}\right) \tag{31}$$

The roots of this polynomial determine the first-order eigenvalue corrections; they are

$$\omega_{1,p\pm}(\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_p) = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{32}$$

Altogether, this means that the effect of multiple acoustic dampers on the eigenfrequencies can be formulated as explicit functions of the perturbation parameters (22). These functions predict the eigenvalue shifts at first order. Only one solution of the implicit unperturbed Helmholtz equation (3) per eigenmode is necessary to establish these functions. Even for degenerate modes, this is the case. This facilitates a much faster assessment of the effect of the various damper parameters.

The Optimization Algorithm

Multiparameter perturbation theory provides the key ingredient for an optimization algorithm to position and tune acoustic dampers. Therefore, the problem can be split into two major steps, namely positioning and tuning of the dampers. An illustration of the scheme is displayed in Fig. 3.

First, the best positioning is computed. The optimization starts with one classical iterative solution of the baseline configuration. Solutions of the Helmholtz equation are the computationally most expensive operations in this algorithm. Evaluations of the explicit multiparameter sensitivity functions (22) are much faster. Thus, assessing all considered positioning patterns with respect to an objective function f in a brute force sensitivity study is feasible when using the base-line solution as input for these asymptotic computations. The value of the objective function is a quality measure of the desired design goals. The function should assess the damping rate of the considered modes and can contain penalty terms accounting for design constraints. Because the argument of the objective function can be expressed as a direct function chain starting with the design parameters ζ_p, continuing with the perturbation parameters ε_p, and ending with eigenvalues $\omega_{(m)}$ of the considered modes it can be explicitly written as

$$f = f(\boldsymbol{\zeta}) \tag{33}$$

The pattern with the best impact to the objective function can then be chosen. It defines a parameter change $\Delta\boldsymbol{\zeta}_{\text{bestpos}}$.

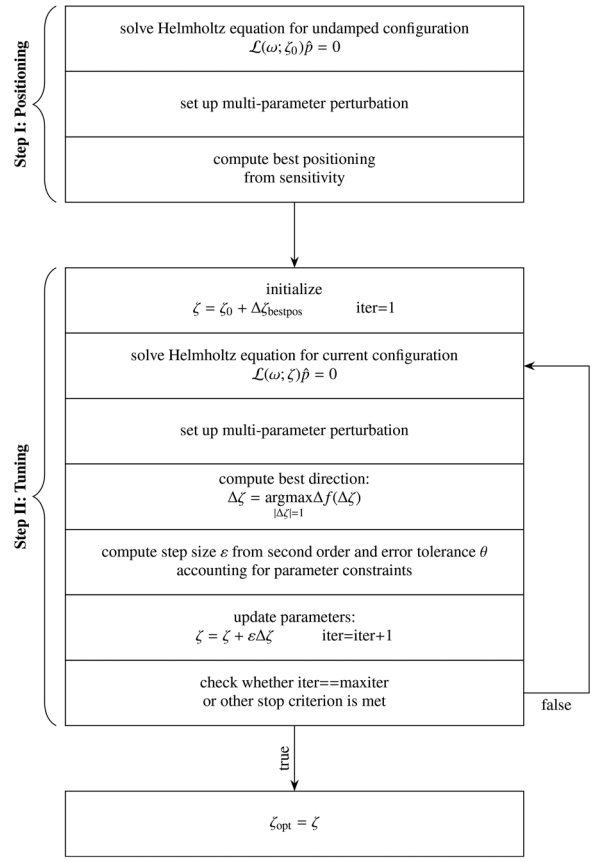

Fig. 3 Schematic illustration of the applied optimization algorithm

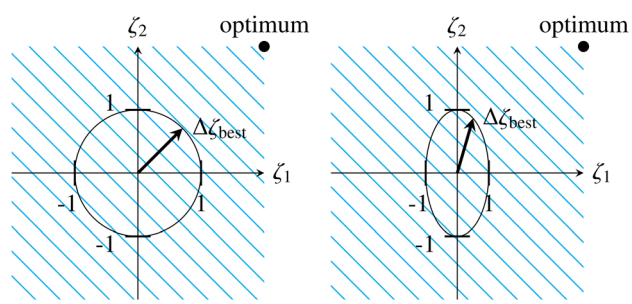

Fig. 4 Illustration of the scaling problem. Cyan lines represent isolines of the objective function. The applied scaling considerably affects the parameter shift.

In the second step, fine tuning of the applied dampers is conducted. The parameter perturbation $\mathcal{L}'$ yielding the best possible local improvement with respect to the objective function f can be found at first order using the multiparameter theory. To achieve this, again only one expensive solution of the Helmholtz equation is required. The results from this solution provide the basis for the perturbative optimization. Therefore, the problem must be formulated as superposition of the individual parameter perturbations

$$\mathcal{L}(\omega) = \mathcal{L}_0(\omega) + \underbrace{\varepsilon \sum_{p=1}^{P} \varepsilon_p \mathcal{L}_p(\omega)}_{\mathcal{L}'(\omega)} \tag{34}$$

It can be interpreted as operator change due to a change $\Delta\boldsymbol{\zeta}$ of the involved parameters $\boldsymbol{\zeta}$ into a certain direction within the parameter

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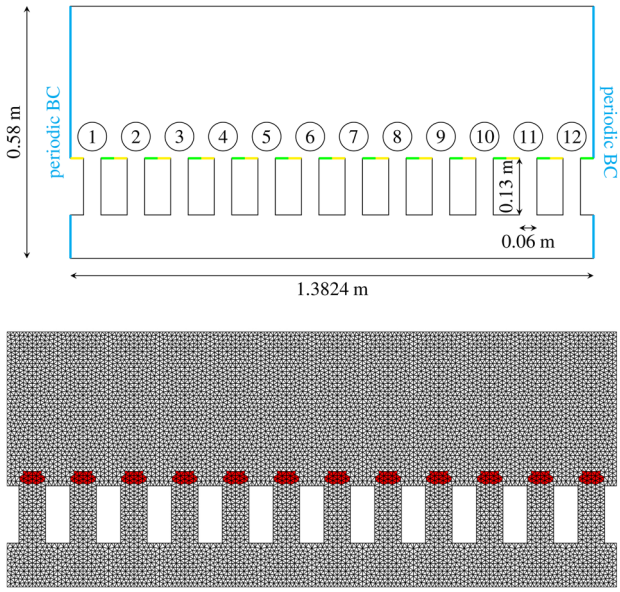

Fig. 5 Illustration of the test case geometry and mesh. The numbers indicate the sector labels. Each sector is equipped with two independent impedance BCs close to the burner outlet (yellow and green lines). Cyan lines indicate periodic boundaries and black lines highlight sound hard boundaries. The red elements of the mesh highlight the domain of heat release.

space. First-order sensitivity analysis facilitates a fast optimization of the involved parameters such that the slope $\Delta f/\Delta\zeta$ of this change is highest with respect to the objective function

$$\Delta\zeta_{\text{best}} = \underset{|\Delta\zeta|=1}{\operatorname{argmax}}\, \Delta f(\Delta\zeta) \tag{35}$$

A practical problem here is the correct scaling of the parameter space. This problem is shortly discussed in the section Scaling the Parameter Space. However, once the best perturbation has been found, the problem remains a single parameter perturbation. The strength ε of this perturbation can be chosen such that the error η of this first-order estimation is below a specified tolerance θ. Second-order single-parameter perturbation theory and the truncation rule can be used to perform this calculation

$$\theta > |\eta| \approx |\epsilon^2\omega_2| \tag{36}$$

$$\Rightarrow \epsilon < \sqrt{\frac{\theta}{|\omega_2|}} \tag{37}$$

Of course, the chosen step size must also account for other parameter constraints, e.g., lower and upper parameter bounds. Iterating this procedure leads to successive improvements of the parameters. This scheme can therefore be characterized as iterative local optimization based on directional search with error-related step size adaption.

Scaling the Parameter Space

The parameters determining the impedance have different physical units. Therefore, an appropriate scaling must be applied when using the optimization algorithm. This is especially necessary to define directions in the parameter space. The applied scaling considerably affects the convergence of the algorithm. Figure 4 illustrates the problem for a two-dimensional parameter space.

The cyan lines represent contour lines of an objective function. The best possible point is to be found in the upper right corner. When defining directions as vectors pointing on the unit disk (unit hyper-sphere for higher dimensions), the applied scaling affects the shape of this disk and therefore the determination of the best possible direction. In particular, the disk may be distorted to an ellipse and the longer semi-axis leads to a possibly overrated direction. For instance, in the right case in Fig. 4, the best parameter change on the disk is not pointing toward the optimum point. The problem is well known in the field of optimization and

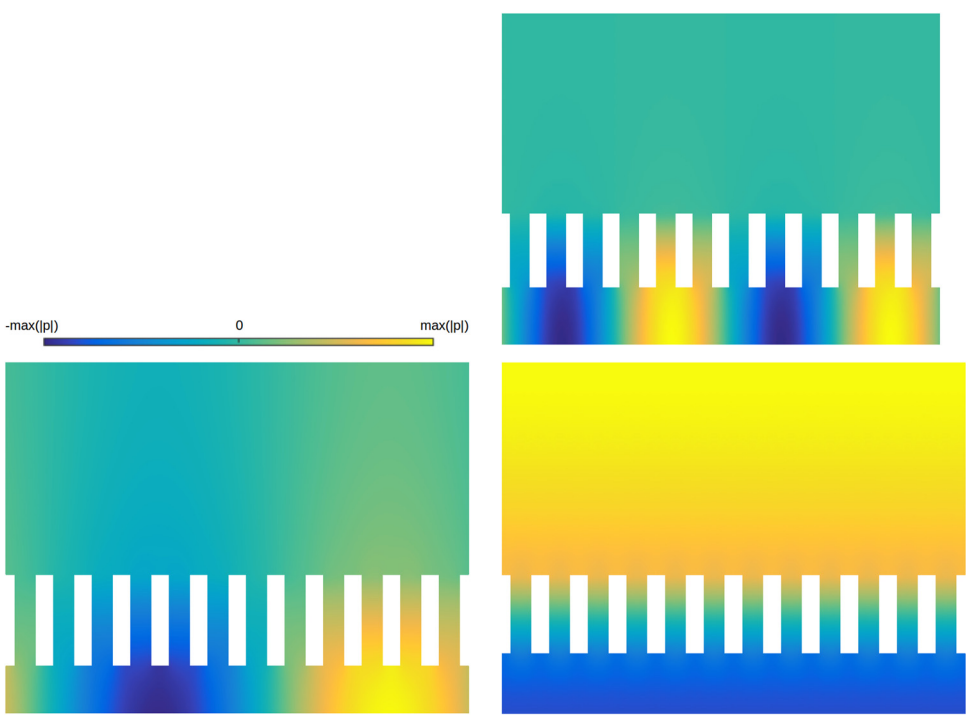

Fig. 6 Mode shapes of the three considered modes. The graphics represent the mode (0) (bottom right), mode (1) (bottom left), and mode (2) (top right). The color shading represents the real part of the complex pressure amplitude $\hat{p}$.

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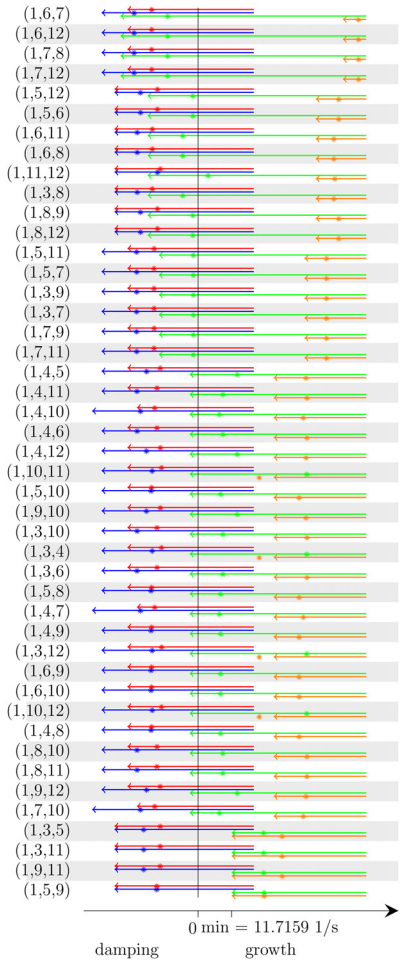

Fig. 7 Comparison of expected (arrows) and actual (stars) growth rate shifts of the first and second considered modes. Orange and green correspond to mode (1); blue and red represent the growth rates of mode (2). The triples indicate the burner segments equipped with dampers. Taking the rotational symmetry of the system into account, all relevant configurations are considered. There are several configurations where the estimate is showing the same low growth rate. This is in accordance with the positioning rule given in Ref. [7].

rotation and rescaling strategies may be applied to overcome this issue; see, for instance, Ref. [17].

Application to a Generic Annular Combustor Model

In order to test the algorithm, a two-dimensional annular combustor model was set up. The model is a simplification of the geometry proposed by Evesque and Polifke [18]. It neglects the radial dimension of the chamber. The geometry and the mesh used for the computations are depicted in Fig. 5.

It features twelve burner tubes connecting a plenum with the actual combustion chamber. Each flame response is modeled in terms of an n-τ-model. The cold gas temperature was set to $T_u = 300\,\mathrm{K}$ and the hot gas temperature to $T_b = 1500\,\mathrm{K}$. The interaction index was set to $n = 0.1$ and the time delay to $\tau = 3\,\mathrm{ms}$. The reference points were located on the center axis of the burner tubes 0.04 m downstream of the burner outlet. Variable impedance boundaries were defined to the left and right of each burner outlet. The ratio of specific heats was set to $\gamma = 1.4$. In total, this allowed to specify 24 independent impedances. For simplicity, a three parameter model was used to model the reduced impedances:

$$\frac{1}{Z_i} = \zeta_{0,i}\frac{\mathrm{i}}{\omega} + \zeta_{1,i} + \zeta_{2,i}\mathrm{i}\omega \tag{38}$$

The acoustic properties of the chamber thus depend on 72 parameters.

The impedance model was proposed by Nicoud et al. in favor of its easy integration into an FEM Helmholtz solver [5]. Its physical interpretation is therefore limited. We use it to test the algorithm. A benefit of the chosen impedance model is that the perturbation operators are linear with respect to the change of the model parameters. Thus, the parameter changes $\Delta\boldsymbol{\zeta}$ can be directly used as perturbation parameters. However, physical impedance models can be described as rational function with analytic nominator and denominator [19]. If such models are used, a linearization has to be applied in order to obtain the perturbation parameters.

Again a finite element method utilizing triangular linear Lagrange elements was used to discretize the problem

$$\underbrace{\left[\mathbf{K} + \sum_{z=1}^{24}\left[\omega\frac{1}{Z_{0,z}}\mathbf{B}_z\right] + \omega^2\mathbf{M} - n\exp(\mathrm{i}\tau\omega)\mathbf{Q}\right]}_{\mathcal{L}_0(\omega)}\mathbf{p} + \sum_{z=1}^{24}\left[\Delta\zeta_{0,z}\mathbf{B}_z + \Delta\zeta_{1,z}\omega\mathbf{B}_z + \Delta\zeta_{2,z}\omega^2\mathbf{B}_z\right] = 0 \tag{39}$$

Initially, all impedance parameters were set to $\zeta_{0,z} = 62.8319$ Hz, $\zeta_{1,z} = 4$, $\zeta_{2,z} = 9.9472$ μs. The optimization algorithm was used to determine and tune the best damper pattern which equips three burner-flame segments with dampers. The minimum damping rate of all considered modes was taken as objective function f. The step size tolerance was set to $\theta = 0.5$ Hz. Three modes were considered, having complex eigenfrequencies of $(\omega_{(0)}/2\pi) = 315.4\,\mathrm{Hz} + 18.5\mathrm{i}\,\mathrm{s}^{-1}$. $(\omega_{(1)}/2\pi) = 350.3\,\mathrm{Hz} - 9.4\mathrm{i}\,\mathrm{s}^{-1}$. and $(\omega_{(2)}/2\pi) = 546.8\,\mathrm{Hz} + 3.1\mathrm{i}\,\mathrm{s}^{-1}$. respectively. The azimuthal mode orders of these modes are 0, 1, and 2. Thus, the second and third modes are two-fold degenerate. The corresponding mode shapes are shown in Fig. 6. 45 different patterns have been tested in a brute force sensitivity study. Some of them are redundant due to the mirror symmetry of the system. However, when considering that rotations do not change the effect of the positioning patterns, all relevant configurations are covered within this study. The results are shown in Fig. 7. The predicted shifts of the growth rates of the modes (1) and (2) are compared with the growth rates obtained when using the classic iterative method to assess the problem.

According to the perturbative approach, the patterns (1,5,9), (1,9,11), (1,3,11), and (1,3,5) feature the lowest growth rate. When considering that the least damped mode in all considered cases is of first azimuthal order, this matches exactly the positioning rule given in Ref. [7]. It states that the best configuration to

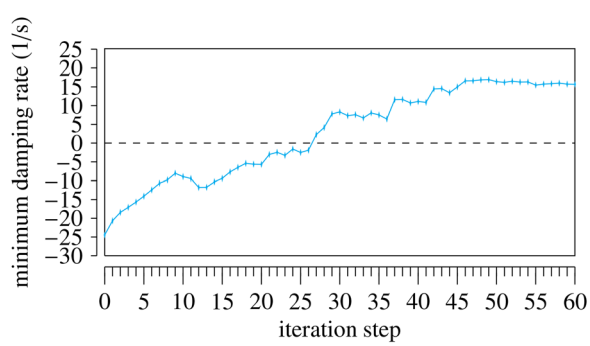

Fig. 8 Evolution of the minimum damping rate for the first 60 optimization iteration steps. The algorithm successfully stabilizes the considered modes. The non-monotonic development may be attributed to poor scaling of the design parameters, a weak step size tolerance, or the non-smoothness of the objective function.

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Table 1 Final damper coefficient after 60 iterations

	Left damper			Right damper		
Sect.	ζ_0	ζ_1	ζ_2	ζ_0	ζ_1	ζ_2
1	541.22 Hz	5.2636	4.8264 ms	541.22 Hz	5.2632	2.6505 ms
3	541.22 Hz	5.2630	6.1987 ms	541.22 Hz	5.2636	4.3310 ms
5	541.22 Hz	5.2630	1.5764 ms	541.22 Hz	5.2632	4.8214 ms

position the dampers is to evenly distribute them within a sector spanning half the circumferential wavelength of the considered mode and rotating any of these dampers by half of this wavelength has no effect. When approximating the effect of the dampers as δ-functions acting in the center of the sectors, these patterns are also in accordance with the C_{2n} -criterion given in Ref. [20], as for all these patterns $C_{2n} = 0$ when considering that the least damped mode is of azimuthal order 1.

Note that for the best configurations, no mode unfolding is predicted for the least damped mode even if the patterns do break the discrete rotational symmetry. Obviously, mode degeneracy is a second-order effect in these cases. This can be seen when considering the results from the iterative solver. Though these accurate results differ from the perturbative predictions, the same configurations were rated as patterns with low growth rate. If the perturbation to the undamped system introduced by the dampers is too strong, the method might fail to predict the correct configurations. A second-order theory together with the truncation rule may be used to estimate the respective limit for the perturbation strength.

Tuning of the damper parameters was conducted for the configuration where the dampers were applied to the sectors 1, 3, and 5. This means that now 18 parameters are considered. The parameter space was rescaled at each iteration according to the parameter shift of the previous step

$$\zeta_{p,\text{scaled},N} = \frac{\zeta_{p,\text{physical},N}}{\Delta\zeta_{p,\text{physical},N-1}} \tag{40}$$

This scaling rule emphasizes those parameters which already had a strong influence to the objective function in the previous step. The evolution of the minimum damping rate of the considered modes over the first 60 optimization iterations is depicted in Fig. 8. It is obvious that the algorithm is capable to improve the damping rate. In particular, all considered modes are stable after the optimization. Some iterations led to a decrease of the minimum damping rate. This may be because

- the step size tolerance was too large
- the applied scaling overrated wrong directions
- the objective function is non-smooth

Further investigations have to clarify the particular reasons. However, after 45 iterations, no significant change of the minimum damping rate occurs. This could be because a local minimum has been reached. However, in the 45th iteration step a branch of mode (2) becomes the least damped mode and the algorithm settings—in particular the applied scaling rule—may not be appropriate to adapt to a new mode shape within a few iterations.

Table 1 shows the final damper coefficients. Most of the parameters have been changed in the same manner. This may be attributed to the chosen scaling rule. However, the parameters significantly deviate from the initial values.

Conclusions and Future Work

An adjoint-based method for the optimum positioning and tuning of multiple acoustic dampers has been derived. As a key aspect, the method utilizes the truncation rule. Application to a model problem indicates that this rule is applicable to solutions of the thermoacoustic Helmholtz equation. The method also rests on a multiparameter formulation of the first-order eigenvalue perturbations. Even if not shown in this paper, it is possible to combine the method with the Bloch wave methodology to exploit the discrete rotational symmetry of the system in order to further accelerate the computations [21]. To account for physical impedance models, linearization of the involved perturbation operators will be necessary. This together with a three-dimensional combustor model will be the topic of future work in order to validate the approach for real combustors.

The optimization algorithm could also be used to optimize other entities than acoustic dampers. For instance, it could be utilized to find the best fuel staging pattern if the effects are available as flame response functions. Moreover, the multiparameter formulation could help calculating uncertainty quantifications like those discussed in Ref. [22]. The main limitations of the framework arise from the fact that the multiparameter perturbation is of first order. Thus, a higher-order perturbation formulation would improve the methodology. For instance, it would allow for global optimization and larger step sizes. Feeding the results from the tuning step back into the positioning step may also lead to improvements of the algorithm. A general application for the presented perturbation theory could also be the acceleration of the iterative algorithms for the solution of the Helmholtz equation. Future work will address these topics.

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